

Running Astroviz Locally

Goal

To run **astroviz** locally and integrate data from a bagfile directly into it.

Setup Steps for Running the Dashboard:

Execute the following commands in sequence if you ONLY want to run the dashboard and use it:

1. Clone the repository:

```
$ git clone https://github.com/mbyx/astroviz.git
```

2. Navigate to the docker directory:

```
$ cd astroviz/docker
```

3. Build the container:

```
$ sh build.sh
```

4. Launch the dashboard:

```
$ sh run.sh dashboard
```

To launch the dashboard with an external data directory:

```
$ sh run.sh dashboard -m path/to/bagfile/folder/
```

To launch the dashboard with a different ROS Domain:

```
$ sh run.sh dashboard -d 40
```

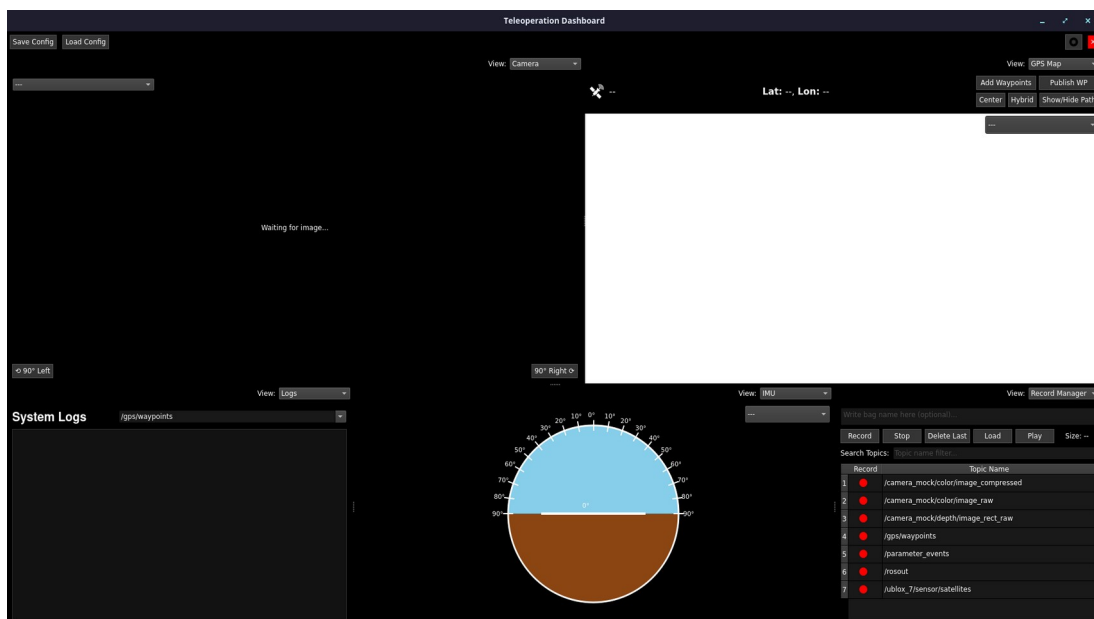


Figure 1: The dashboard running without any data.

Setup Steps for Developing:

Execute the following commands in sequence if you want to start developing new features for the dashboard:

5. Clone the repository:

```
$ git clone https://github.com/mbyx/astroviz.git
```

6. Navigate to the docker directory:

```
$ cd astroviz/docker
```

7. Build the container:

```
$ sh build.sh
```

The steps diverge depending on if you want to:

- Run directly from the terminal:

8. Launch the container:

```
$ sh run.sh
```

9. Build the workspace (Inside the Docker shell):

```
$ colcon build
```

10. Run the dashboard (Inside the Docker shell):

```
$ ros2 launch astroviz astroviz.launch.py
```

- Open the workspace in a DevContainer in VSCode:

11. Launch VSCode in the astroviz/ directory:

```
$ code .
```

12. VSCode will prompt you to reopen in the DevContainer, say yes.

13. Create a new terminal in VSCode and run the following:

```
$ colcon build
```

```
$ ros2 launch astroviz astroviz.launch.py
```